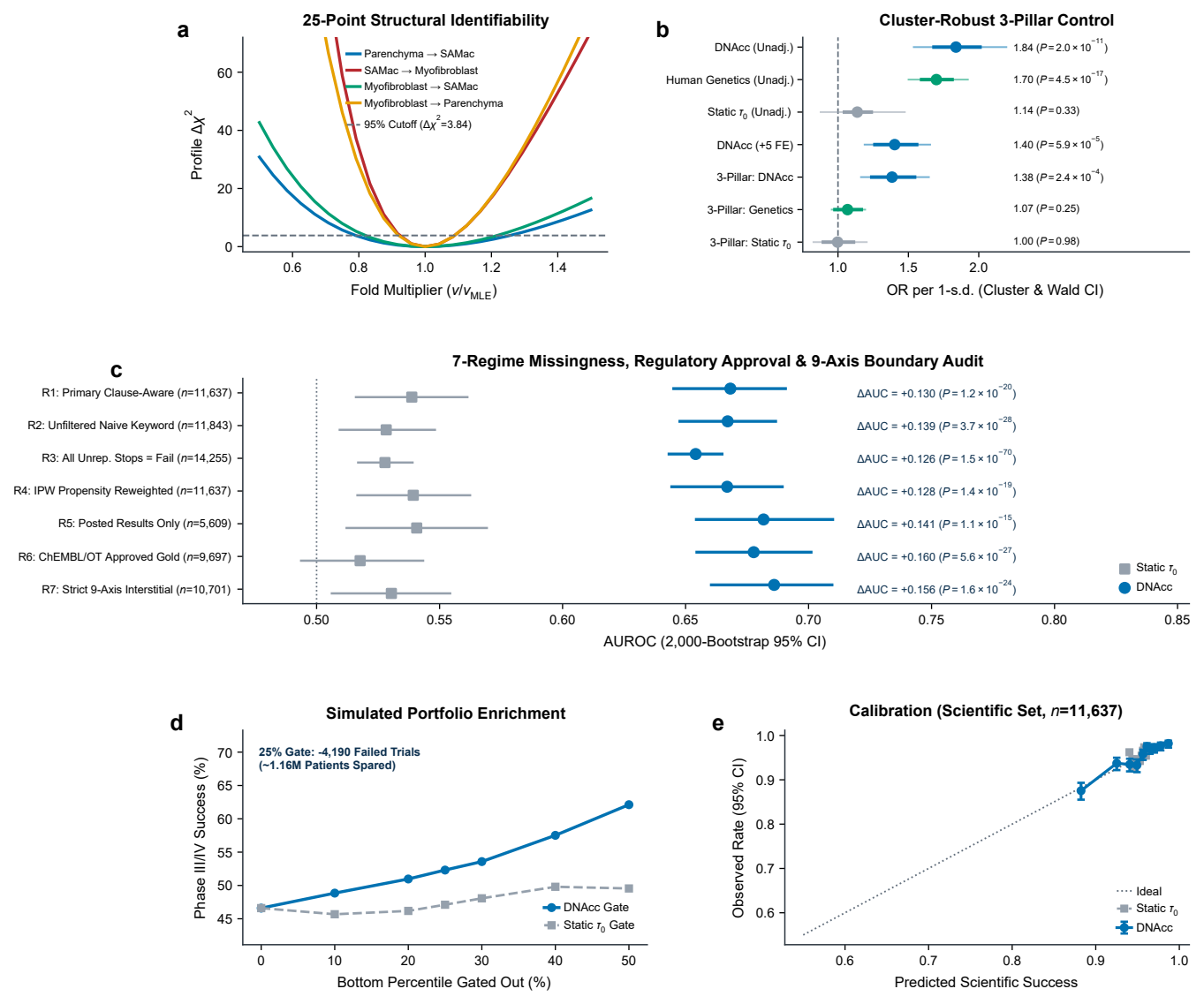

a
25-Point Structural Identifiability
Parenchyma → SAMac
SAMac → Myofibroblast
Myofibroblast → SAMac
Myofibroblast → Parenchyma
95% Cutoff (Δχ²=3.84)
Profile Δχ²
Fold Multiplier (v/v_MLE)
b
Cluster-Robust 3-Pillar Control
DNAcc (Unadj.)
Human Genetics (Unadj.)
Static τ0 (Unadj.)
DNAcc (+5 FE)
3-Pillar: DNAcc
3-Pillar: Genetics
3-Pillar: Static τ0
1.84 (P = 2.0 × 10⁻¹¹)
1.70 (P = 4.5 × 10⁻¹⁷)
1.14 (P = 0.33)
1.40 (P = 5.9 × 10⁻⁵)
1.38 (P = 2.4 × 10⁻⁴)
1.07 (P = 0.25)
1.00 (P = 0.98)
OR per 1-s.d. (Cluster & Wald CI)
c
7-Regime Missingness, Regulatory Approval & 9-Axis Boundary Audit
R1: Primary Clause-Aware (n=11,637)
R2: Unfiltered Naive Keyword (n=11,843)
R3: All Unrep. Stops = Fail (n=14,255)
R4: IPW Propensity Reweighted (n=11,637)
R5: Posted Results Only (n=5,609)
R6: ChEMBL/OT Approved Gold (n=9,697)
R7: Strict 9-Axis Interstitial (n=10,701)
ΔAUC = +0.130 (P = 1.2 × 10⁻²⁰)
ΔAUC = +0.139 (P = 3.7 × 10⁻²⁸)
ΔAUC = +0.126 (P = 1.5 × 10⁻⁷⁰)
ΔAUC = +0.128 (P = 1.4 × 10⁻¹⁹)
ΔAUC = +0.141 (P = 1.1 × 10⁻¹⁵)
ΔAUC = +0.160 (P = 5.6 × 10⁻²⁷)
ΔAUC = +0.156 (P = 1.6 × 10⁻²⁴)
Static τ0
DNAcc
AUROC (2,000-Bootstrap 95% CI)
d
Simulated Portfolio Enrichment
25% Gate: -4,190 Failed Trials
(~1.16M Patients Spared)
Phase III/IV Success (%)
DNAcc Gate
Static τ0 Gate
Bottom Percentile Gated Out (%)
e
Calibration (Scientific Set, n=11,637)
Observed Rate (95% CI)
Ideal
Static τ0
DNAcc
Predicted Scientific Success